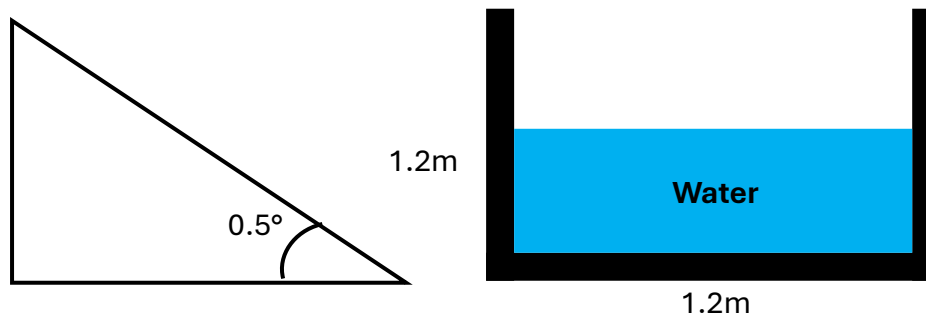


NAME: CALSEÑA, CIENE MART P.	DRY LAB 5	DATE:
HYDRAU LAB 2 – PET 1 – GROUP 2	OPEN CHANNEL	RATING:

A rectangular channel 2.4 m wide has a slope of 0.5° and a water depth of 1.2 m.
Find:

- Q if $n = 0.012$,
- Froude no.,
- Crit. Depth,
- Total Specific Energy, &
- Type of Flow



Solution:

a) Discharge (Q)

$$S = \tan(0.5)$$

- Area (A): $A = (2.4)(1.2) = 2.88 \text{ m}^2$
- Wetted Perimeter (P): $P = 2.4 + 2(1.2) = 4.8 \text{ m}$
- Hydraulic Radius (R):

$$R = \frac{(2.4)(1.2)}{2.4 + 2(1.2)} = 0.6 \text{ m}$$

- Discharge (Q):

$$Q = \frac{AR^{2/3}S^{1/2}}{n}$$

$$Q = \frac{2.88(0.6)^{2/3}(\tan(0.5))^{1/2}}{0.012}$$

$$Q = 15.9493 \text{ m}^3/\text{s}$$

b) Froude Number (F)

- Velocity (V):

$$V = \frac{Q}{A} = \frac{15.9493}{2.88} = \mathbf{5.5380 \text{ m/s}}$$

- Froude Number (F):

$$F = \frac{V}{\sqrt{gd}}$$

$$F = \frac{5.5380}{\sqrt{9.81(1.2)}} = \mathbf{1.6141}$$

c) Critical Depth (D_c)

- Specific Discharge (q):

$$q = \frac{Q}{b} = \frac{15.9493}{2.4} = 6.6456 \text{ m}^2/\text{s}$$

- Critical Depth (D_c):

$$D_c = \sqrt[3]{\frac{q^2}{g}}$$

$$D_c = \sqrt[3]{\frac{(6.6456)^2}{9.81}} = \mathbf{1.6512 \text{ m}}$$

d) Specific Energy (E)

$$E = \frac{V^2}{2g} + d$$

$$E = \frac{(5.5380)^2}{2(9.81)} + 1.2 = \mathbf{2.7632 \text{ m}}$$

e) Type of Flow

Since $F = 1.6141$, and $F > 1$, the flow is **Supercritical or Rapid**.